

Fine-tuning BERT on SST-2 and MRPC

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Abstract

This report presents the fine-tuning and evaluation of a compact BERT model, `prajjwal1/bert-mini`, on two GLUE benchmark tasks: SST-2 for sentiment classification and MRPC for paraphrase identification. The same training configuration was applied to both tasks, and model quality was examined through validation metrics together with representative correct and incorrect predictions drawn from the validation sets.

The quantitative results show that the model achieved stronger performance on SST-2 than on MRPC. After two epochs of fine-tuning, SST-2 reached a validation accuracy of 0.8567, whereas MRPC reached a validation accuracy of 0.7279 and an F1 score of 0.8268. Qualitative analysis shows that the model performs reliably when sentiment is expressed through explicit polarity-bearing words and when sentence pairs preserve nearly identical propositions. In contrast, errors arise more often when sentiment is implicit or comparative, or when two sentence pairs share substantial lexical overlap while differing in the event or relation they express.

1 Introduction

BERT-based models provide contextualized representations that are well suited to downstream natural language understanding tasks. Among the GLUE benchmark tasks, SST-2 and MRPC offer two complementary evaluation settings: SST-2 measures binary sentiment classification for individual sentences, whereas MRPC measures semantic equivalence between sentence pairs. Evaluating the same compact transformer on both tasks makes it possible to compare performance on explicit lexical polarity judgments and on finer-grained semantic matching.

This report examines fine-tuning of `prajjwal1/bert-mini` on SST-2 and MRPC using a shared training configuration. The analysis includes both validation metrics and qualitative inspection of representative predictions. Together, these results provide a compact view of where the model performs reliably and where its errors reveal limitations in semantic interpretation.

2 Models and Experimental Setup

2.1 Model and Data

Experiments were conducted using the scripts `sst2.py` and `mrpc.py`. The model used in both settings was the pre-trained checkpoint `prajjwal1/bert-mini`, paired with the

`bert-base-uncased` tokenizer. SST-2 was treated as a single-sentence binary classification task, whereas MRPC was treated as a sentence-pair binary classification task.

Using the same compact checkpoint across both tasks makes the resulting comparison easier to interpret. Differences in validation behavior can therefore be attributed primarily to task characteristics rather than to changes in model architecture.

2.2 Fine-tuning Configuration

For each task, the dataset and pre-trained checkpoint were loaded, the input text was tokenized, the model was fine-tuned for two epochs, and validation performance was measured at the end of each epoch. The fine-tuned checkpoints were then saved locally and reused for later evaluation on validation examples.

The experimental configuration was as follows:

- Pre-trained model: `prajjwal1/bert-mini`
- Tokenizer: `bert-base-uncased`
- Number of epochs: 2
- Learning rate: 3×10^{-5}
- Batch size: 32 for both training and evaluation

2.3 Execution Evidence

The training code files were run on an NVIDIA RTX 3090 GPU (24 GB). Figures 1 and 2 document the execution outputs for MRPC and SST-2, respectively. These outputs provide direct evidence that fine-tuning, validation, and prediction generation completed successfully for both tasks.

```

root@autodl-container-90ec41be83-76ac87dd:~/autodl-tmp# python3 mrpc.py
Loading weights: 100% | 71/71 [00:00:00, 23877.13it/s]
[transformers] BertForSequenceClassification LOAD REPORT from: prajjwal1/bert-mini
Key | Status |
+---+---+
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.predictions.decoder.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.decoder.bias | UNEXPECTED |
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |
Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downst
ream task.
{'eval_loss': '0.5676', 'eval_accuracy': '0.6961', 'eval_f1': '0.8127', 'eval_runtime': '0.1405', 'eval_samples_per_secon
d': '2904', 'eval_steps_per_second': '92.51', 'epoch': '1'}

Writing model shards: 100% | 1/1 [0
0:00:00:00, 11.26it/s]
{'eval_loss': '0.5456', 'eval_accuracy': '0.7279', 'eval_f1': '0.8268', 'eval_runtime': '0.1054', 'eval_samples_per_secon
d': '3872', 'eval_steps_per_second': '123.4', 'epoch': '2'}

{'train_runtime': '7.928', 'train_samples_per_second': '925.4', 'train_steps_per_second': '29.01', 'train_loss': '0.5783'
, 'epoch': '2'}
100% | 230/230 [0
0:07:00:00, 28.98it/s]

Saving model to bert-for-mrpc
Writing model shards: 100% | 1/1 [0
0:00:00:00, 12.30it/s]
Loading weights: 100% | 73/73 [00:
00:00:00, 7986.44it/s]

Practical task examples:

I wanted to bring the most beautiful people into the most beautiful building , he said Sunday inside the Grand Central co
ncourse . " I wanted to bring the most beautiful people into the most beautiful building , " Tunick said Sunday .

```

Figure 1: Execution output for mrpc.py.

```

root@autodl-container-90ec41be83-76ac87dd:~/autodl-tmp# python3 sst2.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and f
aster downloads.
Loading weights: 100% | 71/71 [00:00:00:00, 13703.09it/s]
[transformers] BertForSequenceClassification LOAD REPORT from: prajjwal1/bert-mini
Key | Status |
+---+---+
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
cls.predictions.decoder.bias | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.predictions.decoder.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.predictions.bias | UNEXPECTED |
classifier.bias | MISSING |
classifier.weight | MISSING |
Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downst
ream task.
[transformers] warmup_ratio is deprecated and will be removed in v5.2. Use 'warmup_steps' instead.
{'loss': '0.5814', 'grad_norm': '4.548', 'learning_rate': '2.938e-05', 'epoch': '0.2375'}

Writing model shards: 100% | 1/1 [00:00:00
:00, 11.80it/s]
{'loss': '0.385', 'grad_norm': '4.249', 'learning_rate': '2.542e-05', 'epoch': '0.4751'}

Writing model shards: 100% | 1/1 [00:00:00
:00, 12.10it/s]
{'loss': '0.3406', 'grad_norm': '4.888', 'learning_rate': '2.146e-05', 'epoch': '0.7126'}

Writing model shards: 100% | 1/1 [00:00:00
:00, 11.96it/s]
41% | 1733/4210 [00:50:01
41% | 1737/4210 [00:50:01:11, 34.39i
41% | 1741/4210 [00:50:01:11, 34.37it/s]
{'loss': '0.3126', 'grad_norm': '5.565', 'learning_rate': '1.751e-05', 'epoch': '0.9501'}

Writing model shards: 100% | 1/1 [00:00:00:00, 11.88it/s]
{'eval_loss': '0.3854', 'eval_accuracy': '0.8417', 'eval_runtime': '0.233', 'eval_samples_per_second': '3742', 'eval_step
s_per_second': '120.2', 'epoch': '1'}
{'loss': '0.281', 'grad_norm': '4.85', 'learning_rate': '1.355e-05', 'epoch': '1.188'}

```

Figure 2: Execution output for sst2.py.

3 Validation Methodology

3.1 SST-2 Evaluation

SST-2 is a binary sentiment classification task in which each example consists of a single sentence labeled as positive or negative. Validation performance is reported primarily through accuracy, which reflects the proportion of sentences assigned the correct sentiment label.

3.2 MRPC Evaluation

MRPC is a binary paraphrase identification task in which each example consists of a pair of sentences labeled as paraphrase or non-paraphrase. In addition to accuracy, F1 is also reported for MRPC, since paraphrase identification is commonly evaluated not only by overall correctness but also by how well positive paraphrase decisions are balanced against misses and false positives.

3.3 Qualitative Analysis Strategy

To complement the aggregate validation scores, representative correct and incorrect predictions were selected from the saved evaluation outputs. This qualitative analysis makes it possible to connect numerical performance to interpretable linguistic patterns, including explicit sentiment cues, implicit evaluative language, lexical overlap, and event-level semantic mismatch.

4 Results

4.1 Overall Quantitative Results

The final validation results after the second epoch are reported in Table 1.

Task	Eval Loss	Accuracy	F1
MRPC	0.5456	0.7279	0.8268
SST-2	0.3763	0.8567	—

Table 1: Final validation performance after the second epoch.

The model achieved higher validation accuracy on SST-2 than on MRPC. On the SST-2 validation set, the model produced 747 correct predictions and 125 incorrect predictions. On the MRPC validation set, it produced 297 correct predictions and 111 incorrect predictions. This pattern is consistent with the greater semantic sensitivity required by paraphrase identification, where small differences in events, relations, or numerical details can alter the label despite substantial lexical overlap.

4.2 Analysis of SST-2 Examples

The SST-2 validation examples indicate that the model is reliable when polarity is expressed through strong lexical cues, but less robust when sentiment is conveyed indirectly

through idiom or comparison.

4.2.1 Correctly Classified Examples

- **“a gorgeous , witty , seductive movie .”** was predicted as label 1 and classified correctly. The sentence contains multiple strongly positive lexical items, and the prediction is therefore consistent with the explicit polarity expressed in the text.
- **“i had to look away - this was god awful .”** was predicted as label 0 and classified correctly. The phrase “god awful” is a highly salient negative cue, while “had to look away” further reinforces an unfavorable evaluation.

4.2.2 Incorrectly Classified Examples

- **“it ’s a cookie-cutter movie , a cut-and-paste job .”** was predicted as label 1, whereas the gold label is 0. Although the sentence is clearly critical, its negativity is expressed through idiomatic descriptions of unoriginality rather than through simple polarity words. The error suggests limited sensitivity to this type of evaluative phrasing.
- **“holden caulfield did it better .”** was predicted as label 1, whereas the gold label is 0. This sentence conveys criticism implicitly through comparison rather than explicit negative wording. The false positive prediction is consistent with difficulty in capturing comparative and context-dependent sentiment.

4.2.3 Additional Discussion

Taken together, these examples suggest that SST-2 performance is strongest when sentiment is encoded by overt polarity-bearing expressions. Errors are more likely when the evaluative meaning depends on idiomatic usage, comparison, or broader discourse interpretation.

4.3 Analysis of MRPC Examples

The MRPC validation examples show that the model performs well when two sentences preserve the same proposition under minor lexical or syntactic variation. Errors arise more often when high surface overlap obscures a difference in the underlying event or relation being asserted.

4.3.1 Correctly Classified Examples

- The sentence pair **“He said the foodservice pie business doesn ’t fit the company ’s long-term growth strategy .”** and **“The foodservice pie business does not fit our long-term growth strategy .”** was predicted as label 1 and classified correctly. The two sentences express the same core proposition with only limited rephrasing, making this a prototypical paraphrase case.
- The pair about **the dollar, the yen, and the Swiss franc** was predicted as label 0 and classified correctly. Despite clear structural similarity and topical overlap, the numerical details differ across the two sentences, so they do not express the same statement.

4.3.2 Incorrectly Classified Examples

- The pair about **Racicot and his wife discussing the war** was predicted as label 1, whereas the gold label is 0. The strong lexical overlap, especially in the clause about using training in the war, may encourage a paraphrase decision, but the two sentences attribute different claims to different speakers.
- The pair about **Terri Schiavo in the hospice** was predicted as label 1, whereas the gold label is 0. Both sentences mention the same person, age, and setting, yet they refer to different events: one concerns an expected death within two weeks, while the other reports that a medical procedure took place. This false positive highlights the difficulty of distinguishing entity overlap from semantic equivalence.

4.3.3 Additional Discussion

These examples indicate that the MRPC model is effective on near-equivalent sentence pairs, but remains vulnerable to false positives when two sentences share substantial lexical and named-entity overlap while differing in the event structure or factual relation that they encode.

5 Conclusion

Fine-tuning `prajjwal1/bert-mini` for two epochs yielded strong validation performance on SST-2 and more moderate performance on MRPC, with accuracies of 0.8567 and 0.7279, respectively, and an MRPC F1 score of 0.8268. The qualitative analysis shows that the model handles explicit sentiment and close paraphrastic variation effectively, but is less reliable on implicit sentiment and on sentence pairs with high topical overlap but different semantic content.

Overall, the results indicate that compact pre-trained transformers can perform competitively on downstream classification tasks, while still exhibiting predictable weaknesses on subtler linguistic phenomena. The contrast between SST-2 and MRPC further shows that task difficulty is shaped not only by label space, but also by the degree to which correct predictions depend on semantic and pragmatic inference beyond surface lexical evidence.